

Assignment 5: Modeling COVID-19

Zeynep Cakez

2024-09-28

Exercise 1

```
days_range <- c(1:250)
```

Exercise 2

```
susceptible <- c(300000000)
infectious <- c(1)
recovered <- c(0)
```

Exercise 3

```
beta <- 0.25
gamma <- 0.125
N <- 300000001

for(day in 2:250){
  susceptible[day] <- susceptible[day - 1] - (beta * susceptible[day - 1] * infectious[day - 1])
  infectious[day] <- infectious[day - 1] + beta * infectious[day - 1] * susceptible[day - 1] - gamma * infectious[day - 1]
  recovered[day] <- recovered[day - 1] + (gamma * infectious[day - 1])
}
```

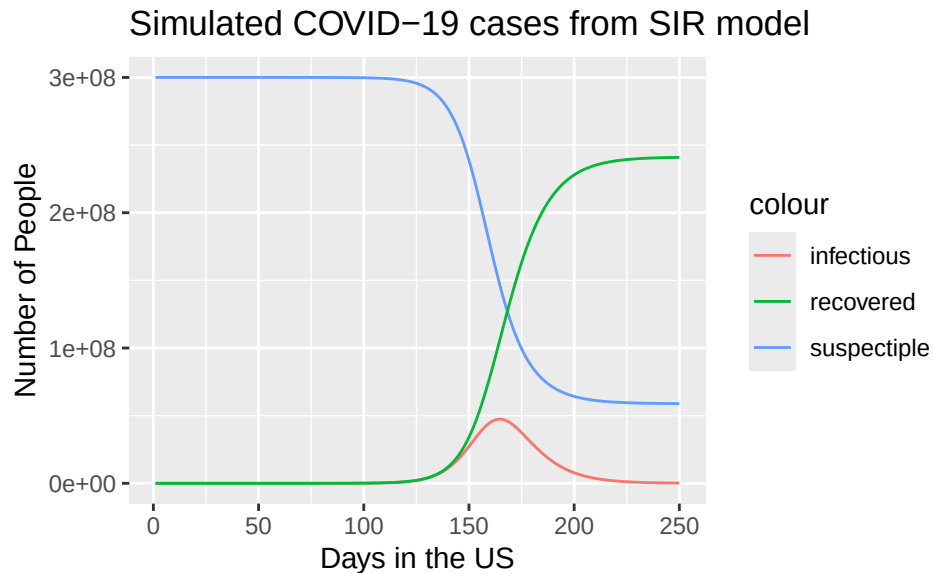
Exercise 4

```
covid_sim <- tibble(
  day = days_range,
  susceptible = susceptible,
  infectious = infectious,
  recovered = recovered
)
```

Exercise 5

```
ggplot(data = covid_sim) +
  geom_line(aes(x = day, y = susceptible, color = "suspectiple")) +
```

```
geom_line(aes(x = day, y = infectious, color = "infectious")) +
geom_line(aes(x = day, y = recovered, color = "recovered")) +
labs(
  title = "Simulated COVID-19 cases from SIR model",
  x = "Days in the US",
  y = "Number of People"
)
```

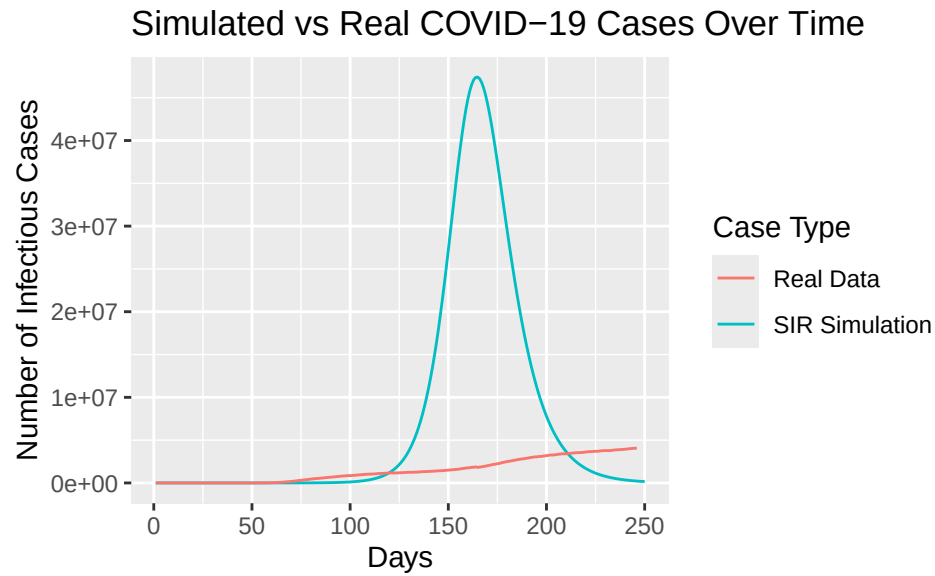


- i. Describe the trends shown in the graph you plotted in the previous exercise. What can you say about the numbers of susceptible, infectious, and recovered people over time, according to our simple model?

the number of susceptible people decrease, while the number recovered people increase. the amount of infectious people increase until about day 165 and the decreases afterwards.

Exercise 6

```
ggplot() +
  geom_line(data = covid_sim, aes(x = day, y = infectious, color = "SIR Simulation")) +
  geom_line(data = covid_real, aes(x = days_in_country, y = current_cases, color = "Real Data")) +
  labs(
    title = "Simulated vs Real COVID-19 Cases Over Time",
    x = "Days",
    y = "Number of Infectious Cases",
    color = "Case Type"
  )
```

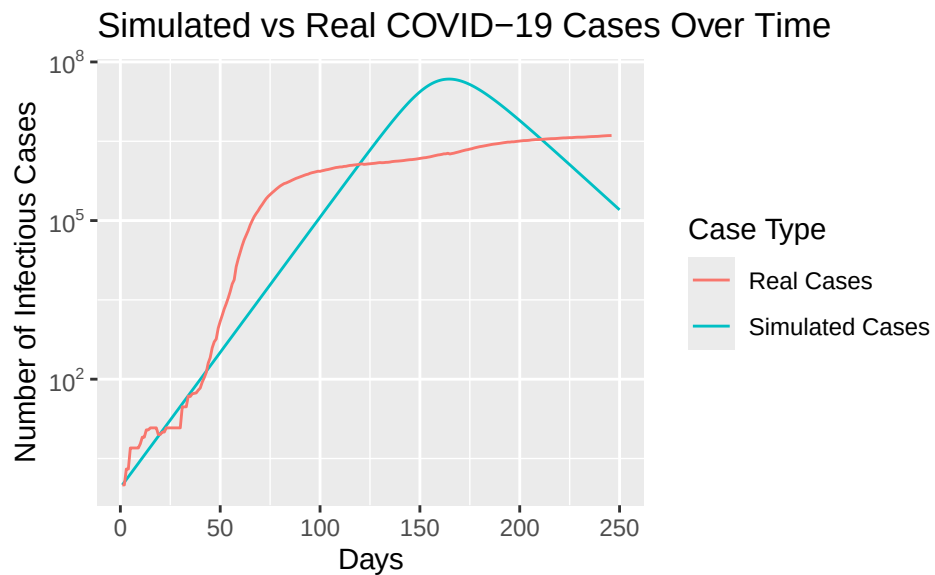


- i. Describe how the SIR model's predictions compare to the real data.

The simulation predicts the peak time number of infection cases way earlier than when it actually happens.

Exercise 7

```
ggplot() +
  geom_line(data = covid_sim, aes(x = day, y = infectious, color = "Simulated Cases")) +
  geom_line(data = covid_real, aes(x = days_in_country, y = current_cases, color = "Real Cases")) +
  labs(
    title = "Simulated vs Real COVID-19 Cases Over Time",
    x = "Days",
    y = "Number of Infectious Cases",
    color = "Case Type"
  ) +
  scale_y_log10(labels = trans_format('log10', math_format(10^.x)))
```



Exercise 8

```

days_range <- 1:250

susceptible <- 300000000
infectious <- 1
recovered <- 0

beta <- 0.3
gamma <- 0.125
N <- 300000001

for (day in 2:250) {
  susceptible[day] <- susceptible[day - 1] - (beta * susceptible[day - 1] * infectious[day - 1]) /
  infectious[day] <- infectious[day - 1] + beta * infectious[day - 1] * susceptible[day - 1] / N -
  recovered[day] <- recovered[day - 1] + (gamma * infectious[day - 1])
}

covid_sim <- tibble(
  day = days_range,
  susceptible = susceptible,
  infectious = infectious,
  recovered = recovered
)

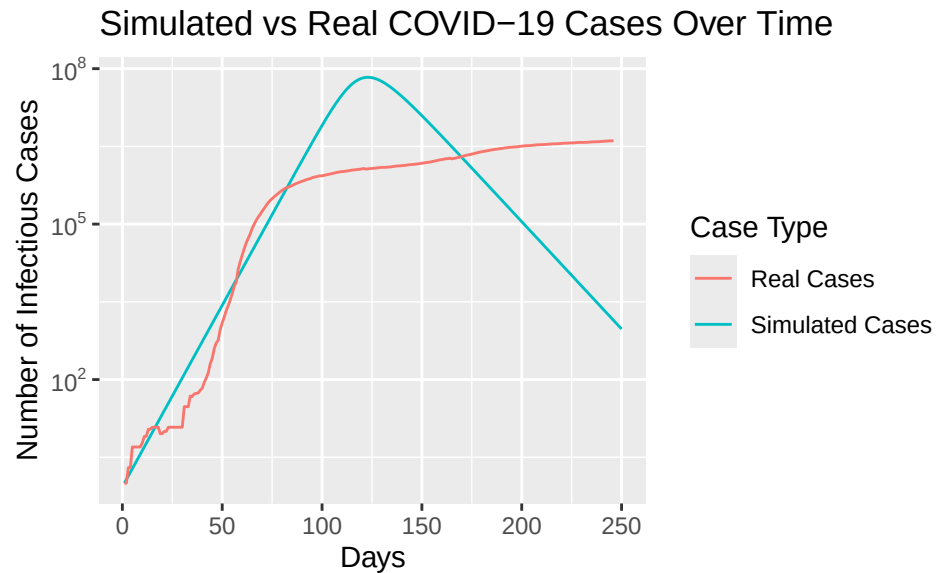
ggplot() +
  geom_line(data = covid_sim, aes(x = day, y = infectious, color = "Simulated Cases")) +
  geom_line(data = covid_real, aes(x = days_in_country, y = current_cases, color = "Real Cases")) +
  labs(
    title = "Simulated vs Real COVID-19 Cases Over Time",
    x = "Days",
  )

```

```

y = "Number of Infectious Cases",
color = "Case Type"
) +
scale_y_log10(labels = trans_format('log10', math_format(10^.x)))

```



```

reproduction <- beta / gamma
reproduction

```

```
## [1] 2.4
```

What is the reproduction number that you have found allows the SIR model to fit the first 60 days of the US COVID-19 pandemic?

2.4

Academic Integrity statement

No tools other than the teachers guidance from Slack was used to complete this assignment.